MIL-C-81104A(AS)
17 February 1971
SUPERSEDING
MIL-C-81104(Weps)
30 June 1964

MILITARY SPECIFICATION

CORD, POLYAMIDE, HIGH TEMPERATURE RESISTANT

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for a high temperature resistant cord for use in anti-G coveralls and related clothing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-T-81	Tags, Shipping and Stock

Military

MIL-C-3131	Cordage, Preparation for Delivery of

STANDARDS

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 4020

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(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Yarn. The yarn used in the manufacture of the cord shall be made from high strength, nonmelting, aromatic polyamide yarn. The yarn shall not carbonize (turn black or become a soft friable char) at a temperature below 427° C when tested as specified in 4.4. The plied yarns shall be twisted only from denier specified in Table I.

TABLE I

PHYSICAL PROPERTIES

CHARACTERISTICS	REQUIREMENT
Yarn	
Denier	200
Ply	3
Twist (turns/inch)	
Spin	7.0 - 9.5
Ply	4 - 6
Picks per inch	26 - 28
No. of carriers and ends per carrier	16/1
Breaking strength, pound, min	100
Elongation, percent, min	30
Length per pound of cord, yard, min	350

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3.2 Color. Unless otherwise specified, the color (shade) shall be as specified by the procuring activity.

3.3 Construction and physical requirements. The finished cord shall conform to the requirements specified in Table I when tested as specified in 4.4.

3.4 Identification ticket. Each holder of cord shall have a ticket attached with not finer than 5-ply cotton string doubled to not less than 8 inches in length. The ticket shall conform to the requirements for Type B, Class 1, Size 4 or 5 of UU-T-81. The ticket shall be made of not less than 20-point paper stock. The color shall be manila, light in intensity to permit easy reading of printed, stamped, typed, or penciled markings. The ticket shall have clipped corners at the end when a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying string. The ticket shall be legibly printed with the following information:

Stock No.
Nomenclature
Date of manufacture
Specification MIL-C-81104A(AS)
Length
Contract No. and date
Supplier's name

3.5 Put-up and length. Unless otherwise specified (see 6.2), the cord shall be put-up in continuous lengths on spools (reels) as specified by the procuring activity.

3.6 Workmanship. The finished cord shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted in accordance with 4.4, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the cord to verify the validity of the certification.

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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in the specification or applicable documents.

4.2.2 Samples for visual examination. The unit of product for this examination shall be one spool (reel). A sample size for the visual examination shall be selected in accordance with MIL-STD-105, Inspection Level II, with an Acceptable Quality Level of 2.5 for major defects and 6.5 for minor defects per 100 units. The defects found shall be identified and classified in accordance with Table II. Defects listed shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted.

TABLE II

CLASSIFICATION OF VISUAL DEFECTS

DEFECT	MAJOR	MINOR
Cut or hole, any.	X	
Abrasion, chafe area or distortion in the orientation of yarn.	X	
Kinks, or unevenly braided resulting in open places, breaks in continuity of braid or soft spots.	X	
Floats, broken or missing ends or picks.	X	
Over 1 knot per 10 linear yards in plied yarns.	X	
One knot per 10 linear yards in plied yarns.		X
Color other than specified.	X	
Put-up other than specified.	X	
Spot or stain clearly visible 1/.		X
Objectionable odor.		X

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TABLE II (Continued)

DEFECT	MAJOR	MINOR
Identification omitted.	X	
Information incorrect, illegible.		X
Ticket not as specified.		X

1/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for length and winding. The sample unit for this examination shall be one spool (reel). The determination of the lot size shall be as specified in 4.2.2. The inspection level shall be Level S-3 and the Acceptable Quality Level shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptable number 1. The lot size shall be the number of spools (reels) in the lot. Defects shall be as listed in 4.2.3.1 and 4.2.3.2. The length on the spool (reel) shall be determined by measuring or by using the following formula:

$$W \times L = \text{Length per spool (reel) in feet}$$

where W = net weight of the spool (reel) in 0.1 pounds

L = lot average of the weight - length relationship (length/pound) as determined in 4.3.1

4.2.3.1 Defects with regard to length. Defects shall be considered to exist if any of the following are determined during inspection:

(a) The length of cord is less than specified or more than 10 percent in excess of the length specified.

(b) Length of cord on spool (reel) is more than 6 feet less than length marked on ticket or label.

4.2.3.2 Defects with regard to winding. Defects shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding or otherwise affecting free unhampered unwinding of cord.

(b) Put-up not as specified.

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4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging and packing requirements of Section 5 of this specification are complied with. The examination shall be in accordance with provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable and as listed in Table IV shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with Table III. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The lot size shall be expressed in units of 1 yard. The sample unit for testing shall be 6 yards of cord.

TABLE III

SAMPLING FOR TESTS

NO. OF UNITS IN LOT	SAMPLE SIZE
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV

TEST METHODS

CHARACTERISTIC	TEST METHOD
Yarn	
Melting point	1534 1/
Denier	4021 1/
Ply	Visual
Twist	
Spin	4052
Ply	4054
Picks per inch	Visual
Breaking strength	4102
Elongation	4102
Length per pound	2/

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1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be accepted for the stated requirement.

2/ See 4.4.1.

* 4.4.1 Determination of length per pound. A specimen of cord shall be put under a tension load equal to 1 percent of the minimum specified breaking strength. A minimum length of 10 feet shall be accurately measured and marked on the specimen to the nearest 1/4 inch. After marking, the tension load shall be removed and length measured shall be cut out of the test specimen and accurately weighed to within plus or minus 1.0 percent of its total weight. Three determinations shall be made per sample unit and the results averaged to the nearest 0.1 foot.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Levels A and C. Cord shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Cord shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. The cord covered by this specification is intended for use in the construction of anti-G coveralls and related clothing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color (shade) required (see 3.2).

(c) Length of roll (see 3.5).

(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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